INCH-POUND

MIL-S-24744(SH)
14 November 1990

MILITARY SPECIFICATION

SCRUBBING MACHINE DECK, SELF PROPELLED, SHIPBOARD

This specification is approved for use by the Naval Sea Systems Command, Department of the Navy, and is available for use by all departments and agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification establishes the performance, design, manufacture, and acceptance requirements for a shipboard, heavy duty, self-propelled, deck scrubbing machine. Only one classification of scrubbing machine is covered by this specification.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed on the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

0-A-548	Antifreeze/Coolant, Engine: Ethylene Glycol, Inhibited, Concentrated
TT-C-490	Cleaning Methods for Ferrous Surfaces and Pretreatments for Organic Coatings

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to Commander, Naval Sea Systems Command, SEA 55Z3, Department of the Navy, Washington, DC 20362-5101 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 3825

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TT-E-489	Enamel, Alkyd, Gloss, Low VOC Content
VV-F-800	Fuel, Oil, Diesel
ZZ-T-381	Tires, Pneumatic, Vehicular (Highway)
ZZ-T-391	Tires, Solid Rubber, and Wheels, Solid Rubber Tire (Industrial)

MILITARY

MIL-S-3950	Switches, Toggle, Environmentally Sealed, General Specification for
MIL-C-5541	Chemical Conversion Coatings on Aluminum and Aluminum Alloys
MIL-T-5624	Turbine Fuel, Aviation, Grades JP-4, JP-5, and JP-5/JP-8ST
MIL-M-8090	Mobility, Towed Aerospace Ground Equipment, General Requirements for
MIL-S-8802	Sealing Compound, Temperature Resistant, Integral Fuel Tanks and Fuel Cell Cavities, High Adhesion
MIL-B-11188	Batteries, Storage: Lead Acid
MIL-T-13867	Treatment, Moisture and Fungus Resistant for Fire Control Electrical and Electronic Instruments and Equipment
MIL-H-17672	Hydraulic Fluid, Petroleum, Inhibited
MIL-P-23377	Primer Coatings: Epoxy, Chemical and Solvent Resistant
DOD-G-24508	Grease, High Performance, Multipurpose (Metric)
DOD-C-24667	Coating System, Nonskid, for Roll or Spray Application (Metric)
MIL-C-38334	Corrosion Removing Compound, Prepaint, for Aircraft Aluminum Surfaces
MIL-C-81706	Chemical Conversion Materials for Coating Aluminum and Aluminum Alloys
MIL-H-83282	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, Nato Code Number H537
MIL-C-83286	Coating, Urethane, Aliphatic Isocyanate, for Aerospace Applications
MS 25243	Lamp, Incandescent PAR-36, Bulb, Screw Terminal Base

STANDARDS

FEDERAL

FED-STD-595	colors

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MIL-STD-130	Identification Marking of U.S. Military Property
MIL-STD-209	Slinging and Tiedown Provisions for Lifting and Tying Down Military Equipment
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-808	Finishes, Materials and Progresses for Corrosion Prevention and Control in Support Equipment
MIL-STD-810	Environmental Test Methods and Engineering Guidelines
MIL-STD-889	Dissimilar Metals

HANDBOOKS

MILITARY

MIL-HDBK-729	Corrosion and Corrosion Prevention Metals

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111 -5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DEPARTMENT OF LABOR (DOL)

Code of Federal Regulations (CFR), Title 49- Transportation

(The Code of Federal Regulations (CFR) is for sale on a subscription basis by the Superintendent of Documents, U.S. Government Printing Office, Washington, DC 20002 When indicated, reprints of certain regulations may be obtained from the Federal agency responsible for issuance thereof.)

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NATIONAL FIRE PROTECTION ASSOCIATION (NFPA)

NFPA No. 70	National Electrical Code - Class 1, Division 2, Group D

(Application for copies should be addressed to the National Fire Protection Association, Batterymarch Park, Quincy, MA 02269.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of DODISS cited in the solicitation. Unless otherwise specified, the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

Boiler and Pressure Vessel Code, Section VIII	Rules for Construction of Pressure Vessels

(Application for copies should be addressed to the American Society of Mechanical Engineers, United Engineering Center, 345 East 47 Street, New York, New York 10017.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

D 3951	Standard Practice for Commercial Packaging; (DOD adopted)
F 1166	Standard Practice for Human Engineering Design for Marine Systems; Equipment and Facilities

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

AMERICAN WELDING SOCIETY (AWS)

D1.1	Structural Welding Code Steel; (DOD adopted)

(Application for copies should be addressed to the American Welding Society, 550 N.W. LeJeune Road, P.O). Box 351040, Miami, FL 33135.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

ARP 1052	Selection Criteria for Reciprocating Prime Movers
ARP 1247	General Requirements for Aerospace Ground Support Equipment; Motorized and Non-Motorized

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ARP 1330	Welding of Structures for Ground Support Equipment
ARP 1801	Measurement of Exterior Sound Level of Specialized Aircraft Ground Support Equipment
J377	Performance of Vehicle Traffic Horns
J512	Automotive Tube Fittings; (DOD adopted)
J514	Hydraulic Tube Fittings; (DOD adopted)
J517	Hydraulic Hose; (DOD adopted)
J524	Seamless Low Carbon Steel Tubing Annealed for Bending and Flaring
J533	Flares for Tubing
J557	High Tension Ignition Cable
J561	Electrical Terminals - Eyelet and Spade Type
J858	Electrical Terminals – Blade Type
J919	Sound Level Measurement –Earthmoving Machinery-Operator
J928	Electrical Terminals - Pin and Receptacle Type
J1127	Battery Cable
J1128	Low Tension Primary Cable
J1401	Road Vehicle – Hydraulic Brake Hose Assemblies for use with Non-petroleum Base Hydraulic Fluids; (DOD adopted)
J1403	Vacuum Brake Hose
J1703	Motor Vehicle Brake Fluid

(Application for copies should be addressed to the Society of Automotive Engineers, Inc. 400 Commonwealth Drive, Warrendale, PA 15096.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. REQUIREMENTS

3.1 First article. When specified (see 6.2). a sample shall be subjected to first article inspection (see 6.5) in accordance with 4.3.

3.2 Item definition. The scrubber shall be a diesel engine powered machine capable of' dispensing a cleaning solution, brush-scrubbing the solution laden deck. pick-up and store solid debris and residual cleaning solution, and with remote extensions, apply high pressure cleaning solution and vacuum-recover the same in areas inaccessible to the main body of the scrubber.

3.2.1 Major component list. The following is a list of all major components which comprise the scrubber:

a. Frame
b. Body
c. Operator station
d. Wheels, brakes, and steering
e. Auxiliary automotive systems
    (1) Electrical system
    (2) Instrument panel
f. Power package and drive train
g. Cleaning system.

3.3 Characteristics.

3.3.1 Performance.

a. Speed, forward: 5 miles per hour (mi/h) minimum, 9 mi/h maximum
b. Speed, reverse: 3 mi/h minimum, 5 mi/h maximum
c. Cleaning rate: 78,000 square feet per hour minimum
d. Cleaning quality: see 4.6.1
e. Capacities:
    (1) Solid debris storage 2.5 cubic feet (ft^3) minimum
    (2) Recovered cleaning solution storage: 30 minutes of scrubbing operation minimum
    (3) Cleaning solution: 30 minutes of scrubbing operation minimum

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(4) Fuel: 6 hours of scrubbing operation minimum

f. Stability: Scrubber shall operate on aircraft carriers under flight deck conditions (simple harmonic motion) as specified in 3.3.5.3.

g. Obstacles:

(1) Scrubber shall negotiate obstructions and depressions maximum 2 inches high or deep.

(2) Scrubbers shall traverse ramps to deck edge elevators and fire door sills on aircraft carriers. The sill requirement is 3 inches of rise with a 4 degree slope on each side of the sill.

h. Service (primary) brakes: Brakes shall hold the scrubber motionless with maximum load whether headed up or down a 15 degree incline. Brakes shall be capable of bringing the loaded scrubber to a full stop within a distance of 24 inches on a dry, level, brush finish concrete surface when moving at a speed of 5 mi/h.

i. Parking (secondary) brake: Parking brake shall hold the scrubber motionless for minimum of 30 seconds on a 11 degree incline and 30 minutes on a 9 degree incline with maximum load whether headed up or down the incline.

j. Duty classification: Intermittent use.

3.3.2 Physical characteristics.

a. Weight: 6000 pounds maximum, unloaded.

b. Dimensions:

(1) Length - 160 inches maximum
(2) Width – 62 inches maximum
(3) Height – 62 inches maximum
(4) Ground clearance -3 inches maximum.

c. Slinging and tiedown provisions: The scrubbers shall be equipped with slinging and tiedown provisions for type IX equipment in accordance with MIL-STD-209, class 1 and 2, or class 3.

3.3.3 Reliability. The scrubber shall operate without failure on an intermittent-use basis during deployment periods of at least 27 weeks at an operating usage rate of at least 60 hours per week. The scrubber shall demonstrate a mean-time-between-critical-failure rate of not less than 4500 calendar hours or 1600 engine-operating hours, whichever occurs first, when the scrubber is operated and maintained in accordance with the manufacturer’s instructions. Critical failure is defined as the inability to operate and clean with the machine as specified in 4.6. Replacement of worn scrubbing or fluid barrier materials does not constitute a critical failure. Any non-repairable component of the cleaning subsystem requiring abrasive contact with the deck surface in order to

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scrub or contain and pick-up cleaning solution and solid debris must demonstrate a mean-time-to-failure rate of not less than 20 engine-operating hours or 60 calendar hours. whichever occurs first, when operated and maintained in accordance with the manufacturer's instructions.

3.3.4 Maintainability.

a. Mean-time-to-repair. 5 hours.

b. Maximum-time-to-repair. 8 hours for any single component exclusive of any logistics delays.

c. Maximum preventive maintenance time: 4 hours per week exclusive of normal servicing of fuel, oil, tire pressure, battery, cooling water, cleaning solution, and removal of accumulated debris and waste water.

3.3.5 Environmental conditions.

3.3.5.1 Temperature. The scrubber shall perform all functions within the temperature range of minus 20 to plus 140 degrees Fahrenheit ($^{\circ}$F) except engine starting which shall be in accordance with 3.5.5.1.2.

3.3.5.2 Wind or jet blast stability. The scrubber shall withstand a wind or jet blast load of 24.4 pounds per square foot (lb/ft^2) (90 mi/h) without overturning. The scrubber must operate safely while subjected to wind loads of 10.9 lb/ft^2 (60 mi/h) from any direction. Wind or jet blast forces are based on:

$$F = C_D \times 0.00252V^2$$

where: F = force in lb/ft^2

C_D = drag coefficient assumed to be 1.2

$0.00252V^2$ = stagnation pressure of dry air at 68 'F and standard atmosphere pressure with a velocity of V mi/h.

3.3.5.3 Flight deck motions. The scrubber shall operate under aircraft carrier flight deck conditions (simple harmonic motion) of plus or minus 9 degrees in roll with a 15.8-second period, and plus or minus 2 degrees in pitch with an 8-second period (sea state 5).

3.3.5.4 Rolling shock The scrubber shall roll over a 2-inch vertical drop at 5 mi/h without experiencing permanent structural deformation or damage to any component.

3.3.5.5 Impact shock. The scrubber shall withstand a 2.5 feet per second (ft/s) impact against an unyielding surface, front and rear, without damage to any component.

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3.3.5.6 Sand and dust. The function of operating parts shall not be impaired by exposure to the sand and dust test, method 510.2 of MIL-STD-810.

3.3.5.7 Salt atmosphere and humidity. The scrubber shall operate satisfactorily during and after exposure to the salt fog test, method 509.2 of MIL-STD-810, and in humidity ranging from 10 to 100 percent.

3.3.5.8 Ice and snow. The scrubber shall start and operate and shall withstand, without permanent deformation or damage to any component, snow or ice loads of up to 15 ib/ft^2.

3.3.5.9 Noise. The maximum average sound level generated by the scrubber, when operated at rated capacity or load, shall not exceed 90 decibels A-weighted (dBA) when measured at the operator's position and at a distance of 15 feet from the equipment at a minimum of 8 positions at 45 degree radials.

3.4 Construction.

3.4.1 Materials, processes, and parts. Materials and parts which are not specified or which are not covered by documents listed herein shall be of the best quality, commercially available, standard parts complying with FED, MIL, or SAE specifications. Processes which are not definitely specified or which are not covered by documents listed herein shall be in accordance with nationally recognized standards (such as AWS, SAE, and so forth). All parts and materials needed to fabricate, assemble, and finish the scrubber shall be furnished by the manufacturer.

3.4.1.1 Recovered materials. Unless otherwise specified herein, all equipment, material, and articles incorporated in the products covered by this specification shall be new and may be fabricated using materials produced from recovered materials to the maximum extent practicable without jeopardizing the intended use. The term "recovered materials means materials which have been collected or recovered from solid waste and reprocessed to become a source of raw materials, as opposed to virgin raw materials. None of the above shall be interpreted to mean that the use of used or rebuilt products is allowed under this specification unless otherwise specified.

3.4.1.2 Asbestos, magnesium and cadmium. The manufacturer shall use materials in the construction of the scrubber and accessories that are free of asbestos, magnesium, and cadmium (see 6.3).

3.4.1.3 Corrosion prevention. Metals exposed to sea water environments shall be corrosion and stress-corrosion resistant or shall be processed to resist corrosion and stress-corrosion. Irrespective of the metals involved, all exposed edges shall be sealed with a suitable sealant material conforming to MIL-S-8802.

3.4.1.3.1 Galvanic corrosion. Joining or coupling of dissimilar metals which are widely separated in the galvanic series should be avoided, if possible. When dissimilar metals must be joined, the juncture surfaces shall be protected in accordance with MIL-STD-889.

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3.4.13.2 Stress-corrosion. The material selection and fabrication processes as specified in MIL-HDBK-729 shall be used to prevent stress-corrosion of metallic parts.

3.4.1.3.3 Electrical and electronic parts. Electrical and electronic parts shall be corrosion resistant or suitably protected in accordance with MIL-STD-454, requirement 15.

3.4.1.3.4 Protective coatings. The scrubber structure and body panels shall be cleaned in accordance with TT-C-490 and corrosion protected as follows:

a. Conditioning (aluminum alloys): Condition with deoxidizer in accordance with MIL-C-38334.

b. Pretreatment (aluminum alloys): Chemical conversion coating in accordance with MIL-C-81706, class 1a, applied in accordance with MIL-C-5541.

c. Pretreatment (ferrous alloys): Chemical conversion coating in accordance with lT-C-490, type I.

d. Primer One coat of MIL-P-23377, type I, class 1, 0.6 to 0.9 roils dry film thickness.

e. Finish: Two coats of MIL-C-83286, FED-STD-595 color yellow number 13538, totalling 1.7 to 2.3 roils dry film thickness. All components used to hold, restrain or protect acid-type storage batteries shall be painted with two coats of commercial-grade acid-proof paint. Instrument and control panels shall be painted with a commercial-grade flat black or wrinkle black finish.

3.4.1.4 Fasteners. Commercial grade fasteners shall be used throughout. All bolted, screwed, or threaded fasteners shall incorporate adequate locking devices. Self-locking nuts are preferred and shall be used on all fasteners in assemblies which are subjected to significant operating vibration. Captive, quick-disconnect fasteners with a positive locking means shall be used on component covers and access plates. Fasteners shall be of corrosion resistant materials or corrosion mated in accordance with MIL-STD-808.

3.4.1.5 Interchangeability. All scrubbers furnished under a specific contract shall be identical to the extent necessary to ensure interchangeability of component parts, assemblies, accessories, and spare parts without modification of parts or the scrubber.

3.4.1.6 Fungus resistance. Materials selected shall be fungus resistant or shall be suitably treated to resist fungus. Electrical and electronic components shall be fungus inert in accordance with MIL-STD-454, requirement 4, or shall be suitably treated to resist fungus in accordance with MIL-T-13867.

3.4.2 Electromagnetic interference. The scrubber shall meet the radiated emissions requirements of MIL-STD-461, UM03, group II vehicles.

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3.4.3 Identification and marking. A unit identification plate in accordance with MIL-STD-130 shall be provided with the following information:

a. Item name and type designation
b. Design activity Commercial and Government Entity (CAGE) number
c. Contract number
d. Unit serial number.

3.4.4 Workmanship. High standards of workmanship and methods of fabrication shall be basic requirements. Sheared or sharp metal edges shall be burred or broken. All exposed metal comers shall be adequately radiused. Welding shall conform to the recommendations of SAE ARP 1330. Welds shall exhibit complete penetration to the root and complete fusion to the base metal. The finished weld shall be free of undercutting, slag inclusions, and cracks. Slag and spatter shall be removed from the weld and surrounding area before painting. Welds on structural members shall not be ground. Non-structural welds may be ground flush where necessary to provide clearance or a smooth surface.

3.4.5 Safety. An electric horn which meets the requirements of SAE J377 shall be provided. The horn shall be supplied with a control button located in the center of the steering wheel.

3.4.5.1 Control pedals. Control pedals shall be constructed and located to permit safe operation if heavy overshoes are worn by the operator. Spacing between the brake pedal and accelerator shall follow standard automotive industry practice. In no case shall they be located so the accelerator can be depressed accidentally while applying the brakes with a booted foot. The use of pedal surface material which inhibits “slippage” of shoes when wet is required.

3.4.5.2 Heat shields. Where required, heat shields or guards shall be installed to protect personnel operating the equipment, or performing routine periodic maintenance on it, against accidental contact with exposed parts which are subject to high operating temperatures. This does not apply to exhaust systems in vehicle engine compartment.

3.4.5.3 Guards. Suitable guards shall be provided for all sprockets, gears, chains, fans, belts, and other moving parts located where operating or maintenance personnel may make accidental contact with them. In general, standard automotive practice is acceptable.

3.4.5.4 Shock protection. Exposure of operating and maintenance personnel to electric shock hazards shall be minimized by the provision of suitable interlocks, grounding means or protective devices.

3.4.5.5 Enclosures. Guards or enclosures shall be provided for all exposed portions of electrical equipment.

3.4.5.6 Relief valves. Relief valves shall be provided in all hydraulic and pneumatic systems to prevent sustained pressures in excess of rated working pressure.

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3.4.5.7 View. The operator shall have clear and unobstructed view 360 degrees around the scrubber when seated in the operator position.

3.4.5.8 Sharp edges. All pinch and shear points, sharp edges, and protruding objects must be eliminated wherever possible and practical. If elimination is not possible, adequate guarding must be provided to prevent damage exposure and injury.

3.4.6 Human engineering. The scrubber shall be easily operated by personnel possessing no special skills and given minimum (not to exceed 16 hours) training. All operator controls and instruments shall be conveniently grouped at one location. Operation shall be easily accomplished while wearing foul weather clothing. The scrubber shall require no organizational or intermediate level maintenance skills beyond those normally provided by an automotive mechanic using standard tools. Access to scrubber components for routine servicing and for maintenance shall be in accordance with ASTM F1166 and the following:

a. Equipment components and systems requiring routine and frequent inspection and maintenance shall be readily accessible. Suitable access doors or removable enclosures shall be provided for this purpose.

b. Access doors, covers, and protective guards shall provide for quick removal or opening. Access holes in protective guards for lubrication are acceptable but shall be held to a minimum number and size.

c. Access panels shall be hinged, pinned, and so forth, to prevent loss from the unit. Large panels of over 4 feet in both height and width which are normally removed only for heavy maintenance; that is, major component overhaul or removal, may be constructed to be removed from the equipment when hinging or pinning is not practical.

d. Hinges shall be located on the forward edge of all vertically hung doors and on the lower edge of all horizontally hung doors. Where possible, at least 8 inches of clearance above the ground shall exist when any door is open. All hinge doors shall have devices to secure them in the closed position so that they will not be blown off by jet blast or ambient winds. Stops or bumpers shall be installed so that the doors, when open, do not mark or scratch the paint work.

e. Major assemblies and components shall be capable of being disconnected and removed from the equipment without the necessity for extensive disassembly of other components. Standard automotive practice is acceptable. Major components should be able to be removed and reinstalled in a period not greater than 8-manhours.

f. Parts and access covers requiring removal or opening for replacement, service, or maintenance shall be fastened with removable fasteners or latching devices.

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g. Weight of removable enclosures, or replaceable units, intended for one-man handling shall be less than 30 pounds. Units exceeding 30 pounds to a maximum of 80 pounds shall be constructed for two-man handling.

h. All components that exceed 80 pounds, or-that exceed 30 pounds if only one man has access to the unit for handling, shall have provisions for attaching lifting or handling devices such as slings, hoists, or forklift tines. The minimum design safety factor of such devices, based on the ultimate strength of the material, shall be 5 to 1.

i. Any vital, unattached parts such as tractor hitch pins and lock pins which might-otherwise be easily lost shall be secured to the equipment with cable or chain of durable size and material.

j. Pressure lubrication fittings shall be provided at all points where heavy loads, close tolerance, relative rotary or linear motion or parts occurs. This will normally include pivots, guides, bearings, and journals except sealed bail and roller bearings. Where access to the fittings is difficult, they shall be brought to a common lubrication panel.

k. Fastener heads and nuts shall be provided with adequate clearance for wrenches or drivers. Supplier shall advise whether metric or U.S. Standard (inches) hardware will be provided.

l. Chain and belt drives shall have provision for adjusting the tension to the manufacturer's specified value. This shall be in the form of take-up screws when needed to produce and hold adequate tension. A positive means to hold the adjustment shall be provided.

m. Operator seating, manual controls, instructions, and visibility shall be in accordance with ASTM F1166 shipboard vehicle requirements.

3.5 Major component characteristics.

3.5.1 Frame.

3.5.1.1 Structure-to-ground clearance. The minimum frame structure-to-ground clearance shall be 6.5 inches when the vehicle is loaded to its normal operating weight.

3.5.1.2 Chassis frame rails. The chassis frame rails shall be lower than any part of the engine, transmission or driveline except the differential housing to protect the components against high crowning damage.

3.5.1.3 Rail loading. The frame rails shall be loaded so that a minimum factor of safety of 3, based on specified yield strength, is maintained under the most severe load condition.

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3.5.1.4 Frame and jack points. No portion of the vehicle frame shall contact the ground with any combination of flat tires. Suitable jack points shall be provided.

3.5.1.5 Stress levels. The stress levels for design shall be based on the total of structural weight plus maximum carried load. Consideration shall be made for anticipated dynamic loads.

3.5.1.6 Structural safety factors. Structural members manufactured of ductile material shall be designed with a minimum factor of safety of 3 based on yield strength.

3.5.1.7 Non-ductile materials. Non-ductile materials shall be constructed for a maximum allowable working stress of one-fourth of the ultimate strength or the material manufacturer's published recommended allowable working stress, whichever is lower.

3.5.1.8 Weld efficiencies. In determining the design factor of safety, weld efficiencies in accordance with AWS D1.1 shall be used.

3.5.1.9 Joint efficiencies. Joint efficiencies shall be included in determination of the factor for bolted connections.

3.5.1.10 Structural rigidity. The unit shall have sufficient structural rigidity that deflections due to load, wind, and motions of working parts do not create interferences, cause malfunctioning of the equipment, or present safety hazards to personnel, aircraft, or the unit itself.

3.5.1.11 Rotary or linear motion parts. Shoulder bolts, bearings, or bushings shall be used when attaching parts having relative rotary or linear motion.

3.5.2 Body panels. All body panels shall be constructed so that permanent damage will not result if the panels are used as stands by personnel. Wherever possible, the top surfaces shall be designed with a slope toward the outside to prevent water accumulation. Suitable drip troughs shall be provided to prevent water from draining from body panels onto the operator or instrument panels. The body panel thickness shall be great enough to provide sufficient structural rigidity so that they will not lose their shape during extended use in a shipboard environment. Panels shall be of sufficient thickness to prevent warpage and distortion when welded in place.

3.5.3 Operator station.

a. The operator’s station shall preclude excessive temperature from heat sources such as engine and exhaust systems. Insulation and ventilation shall be provided, if necessary, to limit compartment temperatures, in any operational mode, to 100 ‘F on a calm, 85°F full-sunlight day.

b. The scrubber is intended for normal operation by one person in the sitting position and shall be equipped with a full seat and backrest adjustable within the range to accommodate the 5th percentile female to the 95th percentile male body dimensions

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as specified in ASTM F1166, while permitting safe and efficient access to all hand and foot controls. The seat or supporting structure shall prevent the transmission of excessive vibration or road shock to the occupant. On non-cab equipped vehicles, a hip guard shall be installed on the outside edge of the scat. Scat material shall be of good commercial marine quality, fire resistant and non-moisture absorbing. Scat dimensions shall be in accordance with ASTM F1166.

c. An adequate support, brace, or floor shall be provided for an operator footrest.

d. Operator station controls, indicators and annunciators shall be as specified in 3.5.5.2.

3.5.4 Wheels, brakes, and steering.

3.5.4.1 Tires. The scrubber shall incorporate commercial pneumatic or solid rubber tires. Pneumatic tires shall be in accordance with ZZ-T-381 and shall be filled with polyurethane foam. The use of solid rubber tires shall be limited to applications where no standard pneumatic tire can be used because of adverse combination of low silhouette, clearance, and load capacity. When used, solid rubber tires shall be of the. pressed-on, or high-profile cushion type with non-skid tread conforming to ZZ-T-391. The solid rubber tire sizes and load capacities shall be as specified in MIL-M-8090.

3.5.4.2 Wheels. Wheels for solid rubber tires shall be industry standard and commercially available. Wheels for solid rubber tires shall have ball or roller bearings, grease seals and dust caps, and shall be removable for wheel replacement.

3.5.4.3 Brakes.

a. The scrubber shall be equipped with brake systems meeting the requirements of DOL CFR, Title 49, applicable to that class of vehicle.

b. The secondary brakes shall be applied by an over-center operating lever actuated by a pushing or pulling motion from the operator's normal position or by a ratchet type foot operated device. Lever mechanism adjustments shall not be available from the operator's station for major brake adjustment; however, minute adjustments to the brake system can be made.

c. The secondary brakes shall be used as an emergency brake in case of failure of the primary brake even though the latter may be comprised of two separate systems. The secondary brakes shall remain applied after initial actuation without further energy input.

d. Hydraulic four-wheel brake systems shall incorporate a tandem master cylinder to split the hydraulic brakes into two separate systems.

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c. All vacuum brake hoses shall conform to SAE J1403.

f. All hydraulic brake hoses shall conform to SAE J1401.

g. Hydraulic brake fluids shall conform to SAE J1703.

h. Hydraulic brake line material shall meet or exceed the requirements of SAE J524. Tubing shall be routed, where possible, so that structural members will provide protection from damage. Tubing shall be firmly anchored to prevent vibration.

i. Brake line tubing shall be formed, bent and flared using hand or powered tools specifically designed for this application. Hand bending without the use of forming took is not acceptable.

j. Brake line connections shall incorporate 45-degree double flared tubing in accordance with SAE J533 mating with inverted flare type automotive tube fittings in accordance with SAE J512.

k. Brake lines shall be routed away from high temperature areas such as engine block and exhaust system.

l. Sleeving used to protect the brake lines from abrasion shall be of non-moisture absorbent material.

3.5.4.4 Steering. The steering system shall provide sufficient mechanical advantage so not to exceed 50 pounds-force steering wheel resistance even when the primary operating mode is power assisted, that is, the operator shall be able to steer the scrubber to a safe stop in the event of a power failure.

3.5.5 Auxiliary automotive systems.

3.5.5.1 Electrical system. The electrical system shall be negative ground and shall consist of an alternator, regulator, 12-volt battery, lights, switches, and necessary wiring. The nominal system voltage shall be 15 volts. The electrical system shall be adequately protected from overloads by discriminate fuses or circuit breakers.

a. Electrical devices including lights, switches, reIays, wiring, and terminals, when located in an area reposed to weather shall be of weatherproof design or protected by weatherproof enclosures.

b. Spark producing electrical components shall be located at least 18 inches above ground level wherever possible. All such components located below this level shall comply with the National Electrical Code requirements for class 1, division 2, group D equipment.

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3.5.5.1.1 Alternator and regulator. The alternator shall be mounted externally on the engine and deliver more amperage than the system draws. The regulator shall maintain voltage levels between 13 and 15 volts.

3.5.5.1.2 Battery. Batteries shall be sized to permit three consecutive, unaided, engine starts at temperatures as low as minus 40 °F with a concurrent wind velocity of 80 mi/h. Batteries shalt be in accordance with MIL-B-11188. The negative pole shall be securely grounded to the vehicle engine and to the vehicle frame.

3.5.5.13 Lighting. The lighting system shall consist of:

a. Two headlights in accordance with MS 25243 and fitted with glass, Aviation Blue lenses similar to FSCM 91729 Part Number 5650

b. Two Marine quality taillights fitted with blue plexiglass lenses.

3.5..5.1.4 Toggle switches. Toggle switches shall be of MIL-S-3950 quality or equal, and rated for the loads controlled. Exposed toggle switches shall be environmentally sealed.

3.5.5.1.5 Wiring and circuit protection.

a. All circuits except starting motors shall have suitable overload protection. Fuses and circuit breakers shall be grouped in convenient locations and suitably marked for size and function. Logical grouping of circuits is anticipated. Headlight circuits shall be independently protected. Protection devices shall be sized to protect wiring and motors from damage due to overload.

b. All wiring shall be in conduit or loomed and shall be routed away from heat sources and fuel lines. Wiring shall be adequately supported to prevent damage from snow and ice buildup, bumping, kinking, and flexing.

c. Common wire splices shall not be used. Connections shall be made using terminal strips and staked lugs or by patent connectors. Terminals shall meet the applicable requirements of SAE J561, J858, and J928.

d. Wiring shall meet the applicable requirements of SAE J557, J1127 and J1128.

e. Any concealed wiring running within inaccessible areas shall be contained in conduit for the length of the run and shall be terminated on a terminal strip at each end of the conduit.

f. Wiring terminals shall be protected by insulating boots or heat shrinkable tubing.

g. Quick disconnect fittings where required shall be MS standard receptacles and plugs.

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h. All electrical connections, including terminal strips and battery terminals protected by an engine compartment hood, shall be provided with covers to prevent accidental contact and short circuiting.

i. Electrical interlock shall be fail-safe design,

j. Each conductor shall be sized to have current carrying capacity as allowed by the National Electrical Code equal to or greater than the capacity of the fuse or circuit breaker provided in its circuit. Optional and add-on components shall be considered in sizing and in the number of conductors provided.

k. Grommets and suitable anti-chafe rnaterial shall be used where wires are required to pass through a firewall or other similar relief or opening which exposes the wire to possible chafing.

1. Each wiring conductor shall be identified by color or number in accordance with a wiring diagram accessibly displayed in the equipment or in an accompanying document.

3.5.5.2 Instruments and controls.

a. Controls and controlling circuits shall be constructed so that any failure within a control or its circuitry will not introduce an unsafe operating condition.

b. Controls shall be grouped and located so as to be convenient to the operator when at his normal operating station but shall be located so as not to permit clothing to catch accidentally on them.

c. Controls shall be in accordance with the shipboard vehicle requirements of ASTM F1166 and this specification.

d. Controls shall permit satisfactory operation when the operator is wearing heavy, arctic-type gloves and overshoes.

e. Controls shall be identified with permanently affixed and non-fading placards.

f. Controls shall be placarded in sharp color contrast in large enough letters or pictograms to be easily read from the operator's position indicating the function and direction of motion of the control.

g. Control panels shall provide easy accessibility of controls and instruments and shall contain all items necessary for the safe operation and control of the equipment. The panel controls and instruments shall be suitably identified and distinctly divided between prime mover and any ancillary equipment.

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h. Instruments and controls exposed to the weather shall be of a ruggedized, weatherproof type and shall be protected from ice and snow accumulations.

i. Accelerator and clutch foot pedal resistance shall be not less than 10 pounds-force and not greater than 20 pounds-force. Brake foot pedals resistance shall be not greater than 180 pounds-force. Hand lever controls, except hand-operated brakes, resistance shall be not less than 2 pounds-force and not greater than 30 pounds-force for movement toward and away from the body, and not greater than 20 pounds-force across the body. Hand-operated brake lever resistance shall be not greater than 50 pounds-force.

j. Controls, except transmission selector levers and brakes, shall move and operate in the direction of travel of the controlled function.

k. ON-OFF switches shall be ON when in the up or forward position.

l. Motion controls, except the transmission selector lever, shall be of the deadman type, returning to neutral position when released.

3.5.6 Power package and drive train.

3.5.6.1 Engine. The engine shall be a liquid-cooled diesel selected from commercial sources in accordance with the criteria of SAE ARP 1052 and the following:

a. Power requirements for driven parts shall not exceed 75 percent of the available horsepower of the engine at operating revolutions per minute (r/rein).

b. Liquid coolant systems shall be rated for maximum engine loads, or at the conditions of maximum intermittent output approved by the engine manufacturer, whichever criterion results in the largest heat transfer capacity.

c. A dry cartridge intake air filter shall be securely mounted in position and shall prevent the emission of flames under backfire conditions. If the engine is to be operated under load in a stationary condition, consideration shall be given to drawing intake air from outside the engine compartment enclosure. Pressure drop across the air filter at rated speed and load shall meet the engine manufacturer's recommendations.

d. A full-flow oil filter shall be installed for the engine oil system.

e. All liquid fillers, drains, dipsticks and filters shall be readily accessible for convenient servicing.

f. Where practical, all driven equipment shall be installed so that the engine is unloaded during cranking (starting).

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g The manufacturer must obtain written approval from the contracting activity to specify use of other than O-A-548, type II. antifreeze.

h. Each engine shall be equipped with an hourmeter and oil pressure switch.

i. During stationary operation when power and speed (r/rein) is not required to be constant, a demand throttle actuated by the control functions will be installed.

j. Diesel engines shall be equipped with a fuel shut-off or an air box shut-off actuated by an emergency kill switch in accordance with SAE ARP 1247.

k. The engine shall be certified for diesel fuel in accordance with W-F-800 and aviation turbine fuel grades JP-4 and JP-5 in accordance with MIL-T-5624.

3.5.6.2 Drive train. The power train gear ratio (transmission, differential, and driving axle) shall be selected to provide optimum conditions for the engine in respect to engine speed, loads and fuel economy in accordance with the engine manufacturer's recommendations. Unless otherwise stated (see 6.2), the vehicle design speed for this replacement shall be considered to be 5 mi/h. Drive shafts and other rotating mechanisms shall be guarded wherever their failure could result in a fuel tank rupture or cause injury to the operator or persons riding on the vehicle.

3.5.6.3 Fuel system.

a. Gravity feed fuel systems shall not be used.

b. Fuel lines shall be of seamless annealed copper or steel tubing. Copper tubing wall thickness shall be not less than 0.035 inch. Steel tubing wall thickness shall be not less than 0.028 inch and shall have a corrosion resistant exterior coating.

c. Fuel lines shall be well supported with adequate clearance from exhaust lines and electrical system parts.

d. Flexible tubing shall be used to absorb vibration and prevent fatigue due to vibration. This is especially applicable for fuel line connections between the frame and flexible mounted engine. Such connections shall be constructed to prevent siphoning in the event of failure of a flexible section.

e. The fuel tank shall have a minimum capacity for 6 hours of scrubbing operation.

f. Fuel tanks shall be located and installed so that any overflow during filling, or any leakage from the tank, lines, or fittings, will not impinge on the engine exhaust system or eletrical equipment, or enter the operator's compartment. Fuel tanks and lines shall be heated so as to be protected from damage in the event of collision.

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g Fuel tanks shall be equipped on the bottom with a drain plug of at least 0.250 inch diameter.

h. Fuel tank fillers shall be accessible from ground level at the outside of the equipment. A-permanent placard shall be installed adjacent to the filler indicating the type of fuel to be used.

i. All fuel tank fill openings shall be equipped with a self-closing fill and attached vent cap. The filler and tank shall provide a fill rate of not less than 5 gallons per minute (gal/min) through the fill cap and filler neck without splash-back for tanks of up to 25 gallon capacity. Larger tanks shall have a fill rate of not less than 20 gal/rein.

j. A non-fluctuating type fuel quantity indicator shall be provided in a location readily visible to the operator. When fuel for more than one system is supplied from a single tank, the tank outlets shall be constructed so that the propulsion engine is the last system affected as the tank approaches empty.

k. The installation of the fuel system shall conform to applicable requirements of DOL 49 CFR 393.67.

3.5.6.4 Exhaust system.

a. Engine back pressures introduced by the complete exhaust system shall not exceed the engine manufacturer's recommendations.

b. The engine exhaust system shall be equipped with flame and spark arresting mufflers.

c. The exhaust system shall be routed clear of all fluid lines and fuel and electrical system components. If routed through areas where leakage of oil, grease, or fuel could occur, the exhaust system shall be shielded from direct contact by such leakage.

d. Flexible exhaust tubing shall not be used.

e. Piping and components shall be installed so that susceptible items such as tires, hoses, and so forth, are not exposed to damaging levels of heat.

f. The exhaust system discharge shall be located so that gases will not enter the operator's position or compartment whether the equipment is stationary or in motion. The discharge shall not be directed toward the deck.

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3.5.6.5 Hydraulic systems. The following requirements shall apply to hydraulic systems other than the chassis brake system:

a. The hydraulic system shall operate over an ambient tcmperature range of minus 20 to plus 140°F.

b. The hydraulic fluid shall be in accordance with MIL-H-17672. A nameplate stating the type of hydraulic fluid used and the total tank capacity shall be installed adjacent to the reservoir filler neck.

c. Maximum allowable flow velocity through any hose, tube, or pipe shall be as follows:

Service	V(ft/s)
Suction	4
Pressure-continuous duty	15
Pressure-intermittent (up to 50 percent duty)	25
Pressure-infrequent (up to 20 percent duty)	40

d. The hydraulic fluid reservoir shall include the following:

(1) Weatherproof breather with 10 micron filtering

(2) A magnetic drain plug incorporated in a sump located at the return end of the tank. The tank shall be arranged so that the sump and drain are at the lowest point.

(3) Full range fluid level indicator

(4) A strainer type filler neck with attached cap

(5) Material and construction conforming to commercial quality and adequately pro-protected against corrosion. Coated tanks are unacceptable.

(6) Items such as strainers, check valves, relief valves, filters, or any other item requiring periodic inspection or repair not located inside the tank, but located outside where they can be serviced easily.

e. A filter canister or a y-type suction strainer of at least 60-mesh Monel or stainless steel screen shall be located between the tank and pump.

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f. The system pumps and components shall be protected by relief valves which have a capacity equal to or greater than pump capacity. Relief valves shall dump directly to tank.

g. Precharge information tags shall be attached adjacent to charge fitting.

h. Hydraulic lines shall be protected and supported from chafing and binding. Hydraulic lines shall be routed so that, where possible, structural members will provide protection. Lines shall be supported so that fittings, tubing, and hoses are separated from engine exhaust systems, and are not subject to damage from heat, external loads, and vibration. If necessary, heat barriers or shields shall be installed. Lines shall be protected from kinking and abrasion.

i. All hydraulic fittings shall be in accordance with SAE J514. If flared, the 37 degree flare with B-nut and sleeve is to be used. Flared copper seats are not to be applied to fittings for sealing purposes.

j. Hydraulic systems shall include devices as necessary to prevent damage to pumps and motors during towing of vehicle at speeds Up to 5 mi/h.

k. Test port locations shall be provided at points in the hydraulic system requiring access for pressure adjustments and trouble-shooting. Each port shall be plugged with a 0.25-inch nominal pipe thread (NPT) plug or closest standard metric equivalent, if applicable.

l. The hydraulic tank filler and breather and lines shall be located away from engine and exhaust system components to prevent oil from splashing onto hot surfaces in the event of overflow, leak, or component failure.

m. Hydraulic hoses shall conform to SAE J517 100R1 through 100R7, as applicable.

3.5.6.6 **Pneumatic systems.** Pressure vessels such as air receivers shall comply with all applicable requirements of ASME Boiler and Pressure Vessel Code, section VIII. Such equipment shall bear an ASME "U" code label and certification. Adequate self-draining filter shall be provided in all pneumatic systems.

3.5.7 Cleaning system. The cleaning system shall:

a. Clean aircraft carrier flight decks, consisting of a non-skid surface in accordance with DOD-C-24667, at a minimum rate of 1300 square feet per minute (ft^2/min) with a cleaning quality that leaves no visual evidence of solid and liquid debris or residual cleaning solution (see 4.6.1).

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b. Dispense cleaning solution and recover and store cleaning solution and liquid debris in sufficient quantity to permit cleaning a 39,000 ft^2 area during a 30-minute period without refilling or draining cleaning solution dispensing and liquid recovery tanks.

c. Pick-up and store 2.5 ft^3 of solid debris consisting of material similar to that described in 4.6.1.

d. Provide a hand held, high-pressure, cleaning solution sprayer capable of cleaning the scrubber, bulkheads, tow tractors, and so forth, at distances up to 20 feet from the scrubber. The sprayer shall have an output nozzle pressure of 700 $\pm$ 100 lb/in^2 gauge and a flow rate of not less than 4 gal/rein. The output nozzle shall be a hand-operated, on-off control type. Hose material shall be industrial-grade, flexible, heavy-duty, abrasion-resistant type. Stowage for the hose and nozzle shall be provided on the scrubber.

e. Vacuum-recover liquid cleaning solution (dispensed by the high-pressure sprayer) at distances up to 20 feet from the scrubber at a rate of 12-gal/min. Hose material shall be industrial-grade, flexible, heavy-duty, abrasion-resistant type. Stowage for the hose and nozzle attachments shall be provided on the scrubber.

f. Cleaning system components that require an abrasion-resulting, friction contact with the non-skid deck in order to clean as specified herein, shall have an operational life in accordance with 3.2.3 and shall not require manual, tool-assisted adjustment or alignment more frequently than once in 10 hours of cleaning operation.

4. QUALTY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified herein or in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program (see 6.2). The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract.

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Sampling inspection, as part of the manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, not does it commit the Government to accept defective material.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection
b. Quality conformance inspection.

4.3 First article inspection. First article inspection shall consist of the examination and tests of 4.5 and 4.6. Visual defects, dimensions not as specified, or any test failure shall be cause for first article inspection.

4.4 Quality conformance inspection. Quality conformance inspection shall consist of the examination and tests of 4.5 and 4.6.

4.4.1 Sampling. Sampling for quality conformance inspection as specified in tables I and II.

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TABLE I. *Sampling for visual examinations.*

Lot size	Sample size
Major defects	
2 – 25	3
26 – 50	5
51 – 90	6
91 – 150	7
151 – 180	10
281 – 500	11
501 – 1200	15
1201 – 3200	18
3201 – 10000	22
10001 – Over	29
Minor defects	
2 – 15	2
16 – 25	3
26 – 90	5
91 – 150	6
151 – 280	7
281 – 500	9
501 – 1200	13
201 – Over	15

Table II. *Sampling for tests.*

Lot size	Sample size
2 – 15	2
26 – 50	3
51 – 90	4
91 – 150	5
151 – 180	6
281 – 500	7
501 – 1200	8
1201 – Over	9

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4.5 Visual examination. Visual examination shall be as specified in table III.

Table III. *Visual examination and classification of defects.*

Examination	Defect	Classification	
		Major	Minor
Finish	Not finished where required	X	
	Type of finish not as specified (blistered, peeled, chipped, or area of no film or thin film)	X	
	Any component of assembly that is painted or otherwise coated where serviceability may be affected. Any area of rust.		X
Dimensions	Dimensions not in accordance with the specified requirements	X	
Materials	Materials not in accordance with the specified requirements	X	
Construction and workmanship, general	Part missing or not specified type or size (except as otherwise indicated herein)	X	
	Fractured, split, punctured, torn, dented, creased, deteriorated, bowed, sprung, malformed, decayed, or otherwise impaired	X	
	Defective or otherwise damaged but not affecting serviceability		X
	Misplaced, loose, or not in proper alignment	X	
	Sharp burr, sliver, or splinter that may injure personnel	X	
	Component not readily accessible for servicing where required		X
Welding or brazing (where required)	Missing, incomplete, burn holes, cracked cracked, fractured, or otherwise not fused	X	
	Slag inclusion, undercut, not smooth and uniform, scale of flux deposits not removed		X

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Table III. Visual inspection and classification of defects - Continued.

Examination	Defect	Classification	
		Major	Minor
Soldering (where applicable)	Missing, not adherent or incomplete incomplete, cold solder joints	X	
	Not clean (flux or flux residue not removed), not smooth (surface not finished neatly), or pin holes in solder	X	
Bolts, nuts, screws, studs and other types of fasteners	Missing, broken, stripped, fractured, or loose	X	
Assembly	Any components not properly assembled or secured	X	
Marking, identification, instruction manual, wiring diagram (where applicable)	Missing, incomplete, not legible, not as specified	X	

4.6 Test methods.

4.6.1 Cleaning. The scrubber shall effectively clean aircraft carrier flight decks, composed of a non-skid surface in accordance with DOD-C-24667. The test surface shall be a flat surface measuring 13 feet wide by 100 feet long. The surface shall be coated with a non-skid coating system in accordance with DOD-C-24667. White and yellow lines, simulating flight deck markings, shall be painted on the test surface. The white and yellow paint shall be in accordance with TT-E-489. The following material shall be distributed over the entire test area:

a. 50 pounds of commercially procured dry sand of the 0.06- to 0.10-inch size
b. 10 gallons aircraft hydraulic-fluid conforming to MIL-H-83282
c. 10 gallons JP-4 or JP-5 aviation turbine fuel conforming to MIL-T-5624
d. 20 pounds multi-purpose grease conforming to DOD-G-24508
e. 100 industrial-grade machine screws, 0.25 inch diameter by 2 inches long
f. 100 industrial-grade hex-head nuts, 0.50-inch size
g. 100 industrial-grade steel washers, 1.50 inch diameter by 0.125 inch thick.

Using a NAVSEA approved flight deck cleaning compound, the scrubber shall clean the entire test surface in a period of not greater than 3 minutes. At the end of this test all solid material shall have been removed from the surface and collected by the scrubber and all free running water and sludge, except that entrapped in cracks or depressions more than 0.25 inch deep, shall have been picked-up and collected by the scrubber. There shall be no evidence of solids or liquids having been pushed off the test surface. Painted lines shall not have been removed or otherwise damaged.

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4.6.2 Vacuum recovery. A container shall be filled with 24 gallons of water and emptied within 2 minutes using only the scrubber vacuum extension hose.

4.6.3 High-pressure sprayer. The output pressure and flow rate shall be measured to determine if the requirements of 3.5.7(d) are met.

4.6.4 Environmental All environmental tests except the watertight test shall be conducted by the contracting activity using Government facilities (see 6.2).

a. Watertight test: The engine cover and instrument panel shall be tested for watertightness. Water shall be pumped through a hose at a nozzle pressure of 15 to 30 lb/in^2. The nozzle diameter shall be not less than 0.5 inch and it shall be not less than 10 feet from the engine cover and instrument panel. The water shall be sprayed directly at the scrubber from all angles for a period of 5-minutes. At the end of the 5-minute period, the engine cover shall be opened and examined for watertightness. The instrument panel shall also be examined for watertightness. Any evidence of water intrusion or damage shall be cause for test failure.

b. Sand and dust: Test method 510.2 as specified in MIL-STD-810

c. Salt atmosphere: Test method 509.2 as specified in MIL-STD-810

c. Cold temperature starting and operation: Test method 502.2 as specified in MIL-STD-810.

4.6.5 Reliability and capacities. The scrubber shall be subjected to tests of the expected life of cleaning components that require and abrasive-type frictional contact with the non-skid test surface; cleaning solution capacity; recovered liquid capacity; and fuel capacity. The scrubber shall be fully loaded with water in the cleaning solution tank and with an amount of fuel not greater than 50 percent of the fuel tank capacity, and operated in the scrubbing mode on the test surface for a minimum period of 30-minutes. During the 30-minute period there shall be no evidence of interruption of cleaning solution flow to the test surface. At the end of the 30-minute period, the test surface shall be free of all dispensed cleaning solution except that entrapped in cracks or depressions more than 0.25 inch deep. After evaluation of the 30-minute of test, the scrubber shall resume scrubbing operation until a minimum total of 3.0 hours of scrubbing have accumulated. During this test phase, the scrubber may be stopped only for periods long enough to refill water in the cleaning solution tank. The scrubber shall not have run out of fuel during the test period nor shall there be any evidence of water remaining on the test surface except that entrapped in cracks or depressions more than 0.25 inches deep. At the end of this test, measurements shall be made of all cleaning system components that require an abrasive-type friction contact with the non-skid surface to calculate the amount of wear incurred. The total life of the worn material shall be extrapolated, factoring in the designed capability to periodically adjust/rotate the material. The resultant calculated life shall be not less than that specified in 3.3.3.

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4.6.6 Stability. Stability shall be evaluated by the following procedures:

a. Place the fully loaded scrubber (with no operator) on a tiltable platform coated with a non-skid surface in accordance with DOD-C-24667. The longitudinal axis shall be parallel to the edge of the platform which will serve as a pivot.

b. Slowly raise the edge opposite the pivot until the platform makes an angle of 9 degrees with the horizontal. Observe the results, then lower the platform.

c. Turn the equipment 15 degrees about its vertical axis, reraise the platform, and observe the results.

d. Repeat step (c) until 24 observations, each at a different angle, have been made.

e. Equipment fails this test when any of the following occurs:

1. Vehicle upsets
2. Parking/emergency brakes do not hold
3. Vehicle slides on platform.

4.6.7 Rolling shock. The scrubber shall pass the following test without permanent material deformation or component damage when loaded to rated capacities:

a. Roll the scrubber along a smooth surface at 5 mi/h over a 2-inch vertical drop onto a hard surface

b. Repeat this operation for a total of two times, once forward and once in reverse.

4.6.8 Impact shock. The scrubber shall be subjected to a 2.5 ft/s impact against an unyielding surface on the front and on the rear without damage to the equipment. Minor structural, or cosmetic damage to bumpers shall not be cause for test failure.

4.6.9 Service brakes. The scrubber service brakes shall be evaluated by the following procedures:

a. Place the fully loaded scrubber on a tiltable platform coated with a non-skid surface in accordance with DC) D-C-24667. The longitudinal axis shall be perpendicular to the edge of the platform which will serve as a pivot. Engage only the service brakes (by means of an operator or other approved method).

b. Slowly raise the edge opposite the pivot until the platform makes an angle of 15 degrees with the horizontal. Observe the results for 1 minute, then lower the platform.

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c. Turn the scrubber 180 degrees about its vertical axis. re-engage only the service brakes, reraise the platform and observe the results for 1 minute.

d. Brakes shall bring the loaded scrubber to a full stop within a distance of 24 inches on a dry-level, brush-finish, concrete surface, when moving at a speed of 5 mi/h.

e. Equipment fails this test when any of the following occurs:

1. Vehicle slides on platform
2. Vehicle stops at a distance exceeding 24 inches.

4.6.10 Parking brake. The scrubber parking brake shall he evaluated by the following procedures:

a. Place the fully loaded scrubber on a tiltable platform coated with a non-skid surface in accordance with DOD-C-24667. The longitudinal axis shall be perpendicular to the edge of the platform which will seine as a pivot. Engage and lock only the parking brake.

b. Slowly raise the edge opposite the pivot until the platform makes an angle of 11 degrees with the horizontal. Observe the results for 30 seconds, then lower the platform.

c. Again raise the edge opposite the pivot until the platform makes an angle of 9 degrees with the horizontal. Observe the results for 30 minutes, then lower the platform.

d. Turn the scrubber 180 degrees about its vertical axis, re-engage and lock only the parking brake, and repeat (b) and (c) above.

e. Equipment fails this test if any evidence of slippage is observed.

4.6.11 Obstacles. The scrubber shall be operated in the scrubbing mode and propelled at 5 mi/h over a fixed obstruction not less than 2 inches high, at least three times. In the non-scrubbing mode, the scrubber shall negotiate (at 5 mi/h) fire door sills, consisting of a 4-degree slope with 3 inches of rise on each side of the sill, at least three times. Damage to any part or the scrubber is cause for test failure.

4.6.12 Noise. Sound level measurements and techniques shall be in accordance with SAE J919 for the operator’s position and in accordance with SAE ARP 1801 for the radial measurements. Sound level in excess of that specified in 3.3.5.9 cause for test failure.

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4.6.13 Design proof load. The scrubber shall be loaded to twice the rated load in such a manner that the load will not be carried by the tires. Evidence of permanent structural or body deformation is cause for test failure.

4.6.14 Electromagnetic compatibility (EMC) tests. The scrubber shall be tested for EMC in accordance with MIL-STD-462 to determine compliance with the group 11. UM03, vehicle requirements of MIL-STD-461.

4.7 Inspection of packaging. Sample packages and packs, and the inspection of the preservation, packing, and marking for shipment and storage shall be in accordance with the requirements of section 5 and the documents specified therein.

5. PACKAGING

(The packaging requirements specified herein apply only for direct Government acquisition. For the extent of applicability of the packaging requirements of referenced documents listed in section 2, see 6.7)

5.1 Commercial packaging. Each scrubber and accompanying attachments shall be preserved, packed, and marked in accordance with ASTM D3951.

5.2 Packaging materials. Asbestos or materials and items containing asbestos shall not be used in the packaging of the scrubber, attachments, and provisioned items.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Extended use The shipboard deck scrubbing machine (scrubber) is intended for use aboard aircraft earners and amphibious assault ships to clean flight and hangar decks.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number, and date of this specification

b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1 and 2.2)

c. When first article inspection is required (see 3.1)

d. When vehicle design speed is other than as specified (see 3.5.6.2)

e. When the watertight environmental test is to be conducted (see 4.6.4).

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6.3 Consideration of data requirements. The following data requirements should be considered when this specification is applied on a contract. The applicable Data Item Descriptions (DID's) should be reviewed in conjunction with the specific acquisition to ensure that only essential data are requested/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure that only essential data are requestcd/provided and that the DID's are tailored to reflect the requirements of the specific acquisition. To ensure correct contractual application of the data requirements, a Contract Data Requirements List (DD Form 1423) must be prepared to obtain the data, except where DOD FAR Supplement 27.475-1 exempts the requirement for a DD Form 1423.

Reference Paragraph	DID Number	DID Title	Suggested Tailoring
3.4.1.2	DI-E-2121	Certificate of Compliance	

The above DIDs were those cleared as of the date of this specification. The current issue of DOD 5010. 12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL), must be researched to ensure that only current, cleared DIDs are cited on the DD Form 1423.

6.4 Technical manuals. The requirements for technical manuals should be considered when this specification is applied on a contract. If technical manuals are required, military specifications and standards that have been cleared and listed in DOD 5010.12-L, Acquisition Management Systems and Data Requirements Control List (AMSDL) must be listed on separate Contract Data Requirements List (DD Form 1423), which is included as an exhibit to the contract. The technical manuals must be acquired under separate contract line item in the contract.

6.5 First article. When first article inspection is required, the contracting officer should provide specific guidance to offerors whether the item should be a preproduction sample, a first article sample, a first production item, a sample selected from the first production items, a standard production item from the contractor's current inventory (see 3.1) and the number of items to be tested as specified in 4.3. The contracting officer should also include specific instructions in acquisition documents regarding arrangements for examinations, approval of first article test results, and disposition of first articles. Invitations for bids should provide that the Government reserves the right to waive the requirements for samples for first article inspection to those bidders offering a product which has been previously acquired or tested by the Government, and that bidders offering such products, who wish to rely on such production or test, must furnish evidence with the bid that prior Government approval is presently appropriate for the pending contract. Bidders should not submit alternate bids unless specifically requested to do so in the solicitation.

6.6 Provisioning. Provisioning Technical Documentation (PTD), spare parts, and repair parts should be furnished as specified in the contract.

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6.6.1 When ordering spare parts or repair parts for the equipment covered by this specification, the contract should state that such spare parts and repair parts should meet the same requirements and quality assurance provisions as the parts used in the manufacture of the equipment. Packaging for such parts should also be specified.

6.7 Sub-contracted material and parts. The packaging requirements of referenced documents listed in section 2 do not apply when material and parts are acquired by the contractor for incorporation into the equipment and lose their separate identity when the equipment is shipped.

6.8 Subject term (key word) listing.

Aircraft foreign object damage
FIight deck scrubbing
High pressure washer
Vacuum-recovery

Preparing activity
Navy - SH
(Project 3825-N194)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

NOTE: This form may not be used to request copies of documents, nor to request waivers, or clarification of requirements on current contracts. Comments submitted on this form do not constitute or imply authorization to waive any portion of the referenced document(s) or to amend contractual requirements.

I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER	2. DOCUMENT DATE (YYMMDD)
	MIL-S-24744(SH)	14 November 1990

3. DOCUMENT TITLE
SCRUBBING MACHINE, DECK, SELF PROPELLED, SHIPBOARD

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME Technical Point of Contact(TPOC): James Brechka PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	b. TELEPHONE (Include Area Code) (1) Commercial TPOC: (215) 897-7317	(2) AUTOVON 442-7317
c. ADDRESS (Include Zip Code) Commander, Naval Sea Systems Command Department of the Navy (SEA 55Z3) Washington, DC 20362-5101	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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